

Acoustic Biodiversity Mapping

A global location-level biodiversity metric from Xeno-canto recordings

Generated 2026-09-28

OBJECTIVE

Derive a single metric that classifies a geographic location as biodiversity 'good' or 'bad', using a global archive of wildlife sound recordings and their metadata.

DATA

- 1,003,444 recordings across 5 continents (Xeno-canto, 1886-2025); 758,198 of them carry 40 acoustic-index columns (16 indices) from a SLURM batch pipeline (compute_indice.py, filtered audio).
- Metadata flattened from 766,747 API records, plus 245,246 older recordings merged in with their own metadata and indices (merged_metadata_all.csv).
- Those merged indices come from different settings (Bioacoustic ~36x ours on the same clips), so their 232,035 geolocated rows count toward richness but NOT index means.

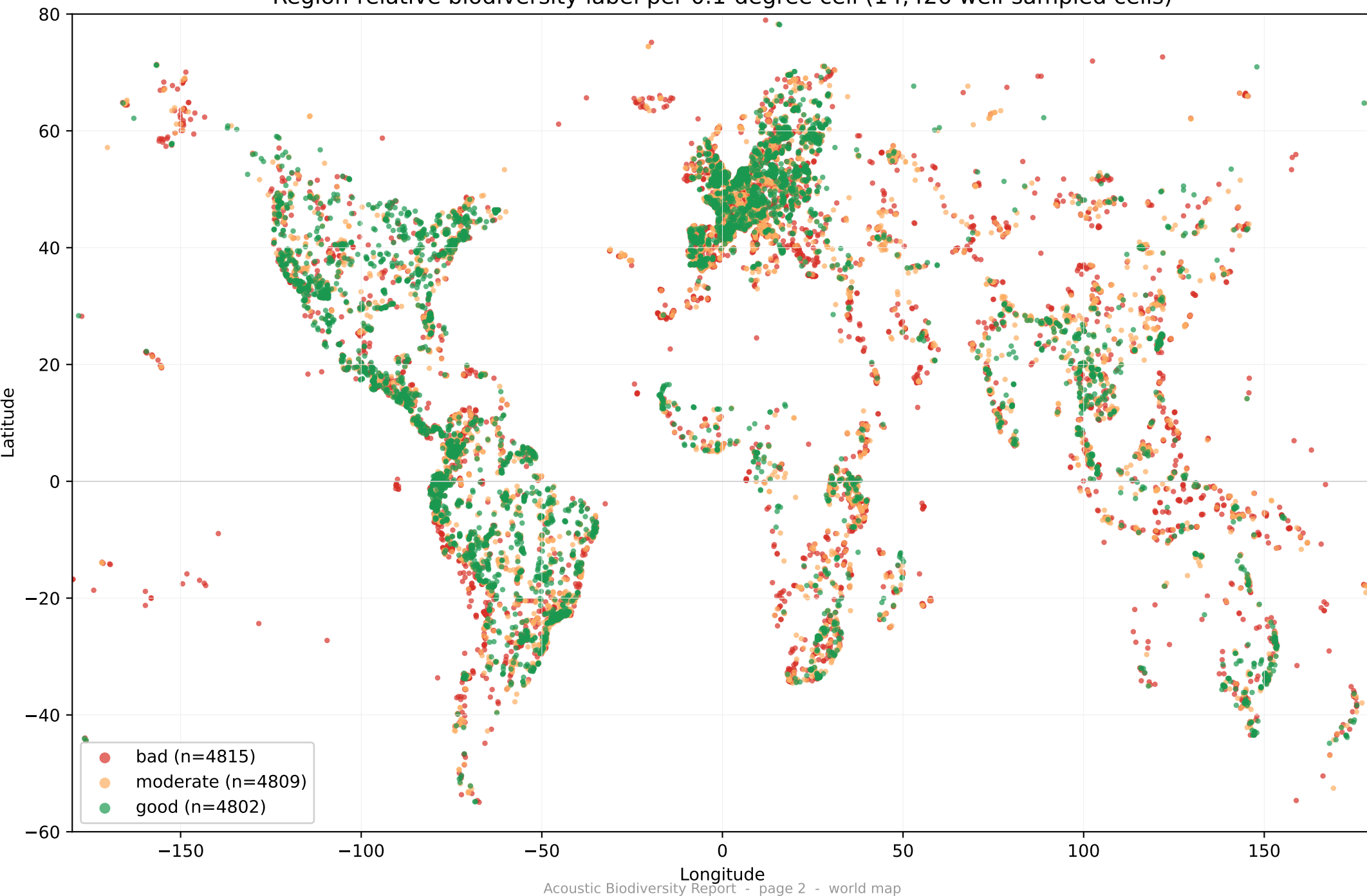
METHOD

1. Aggregate all recordings into 0.1-degree (~11 km) grid cells -> 49,638 cells.
2. Per cell, compute species richness from recorded species (genus+species and the 'also' co-occurring-species field), corrected for sampling effort by Hurlbert rarefaction to 10 recordings (S_{rare10}).
3. Test whether the acoustic indices predict richness (validation against ground truth).
4. Label each well-sampled cell good / moderate / bad relative to its 10-degree latitude band (region-relative tertiles).

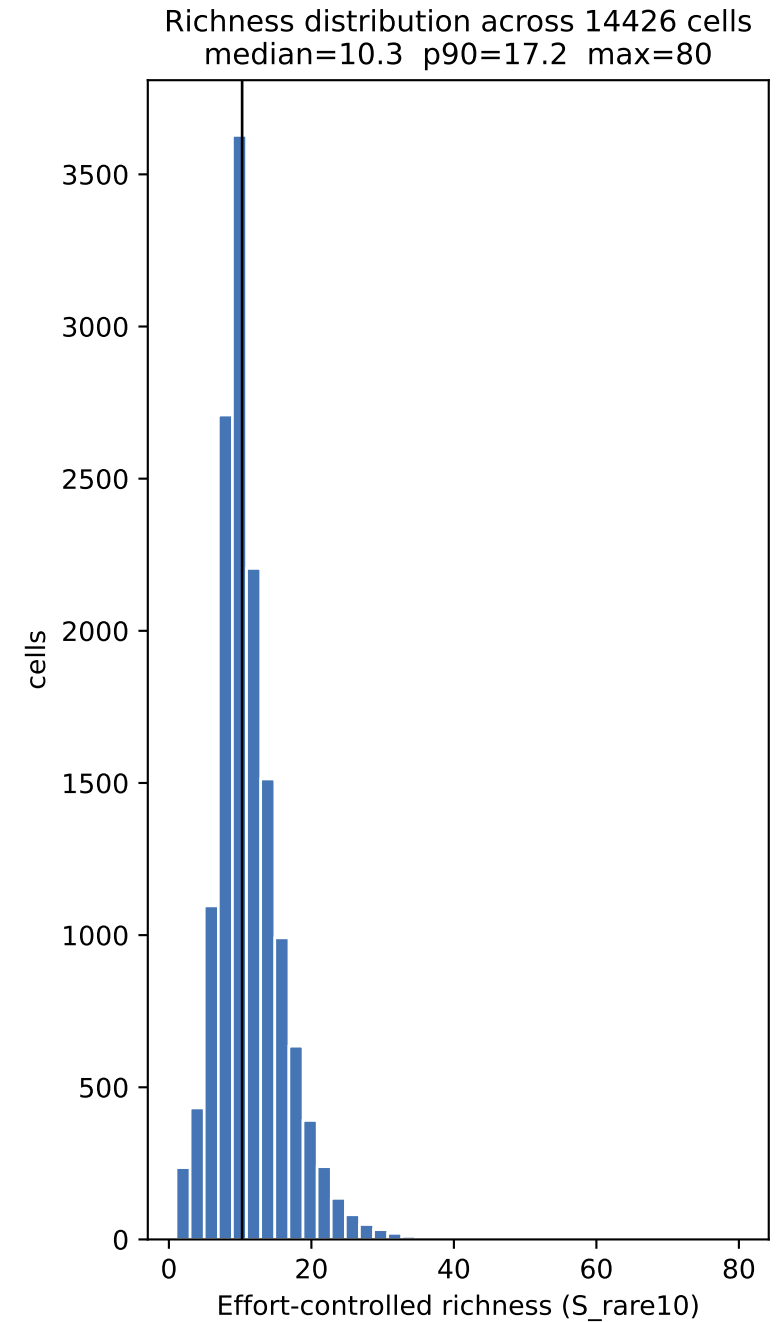
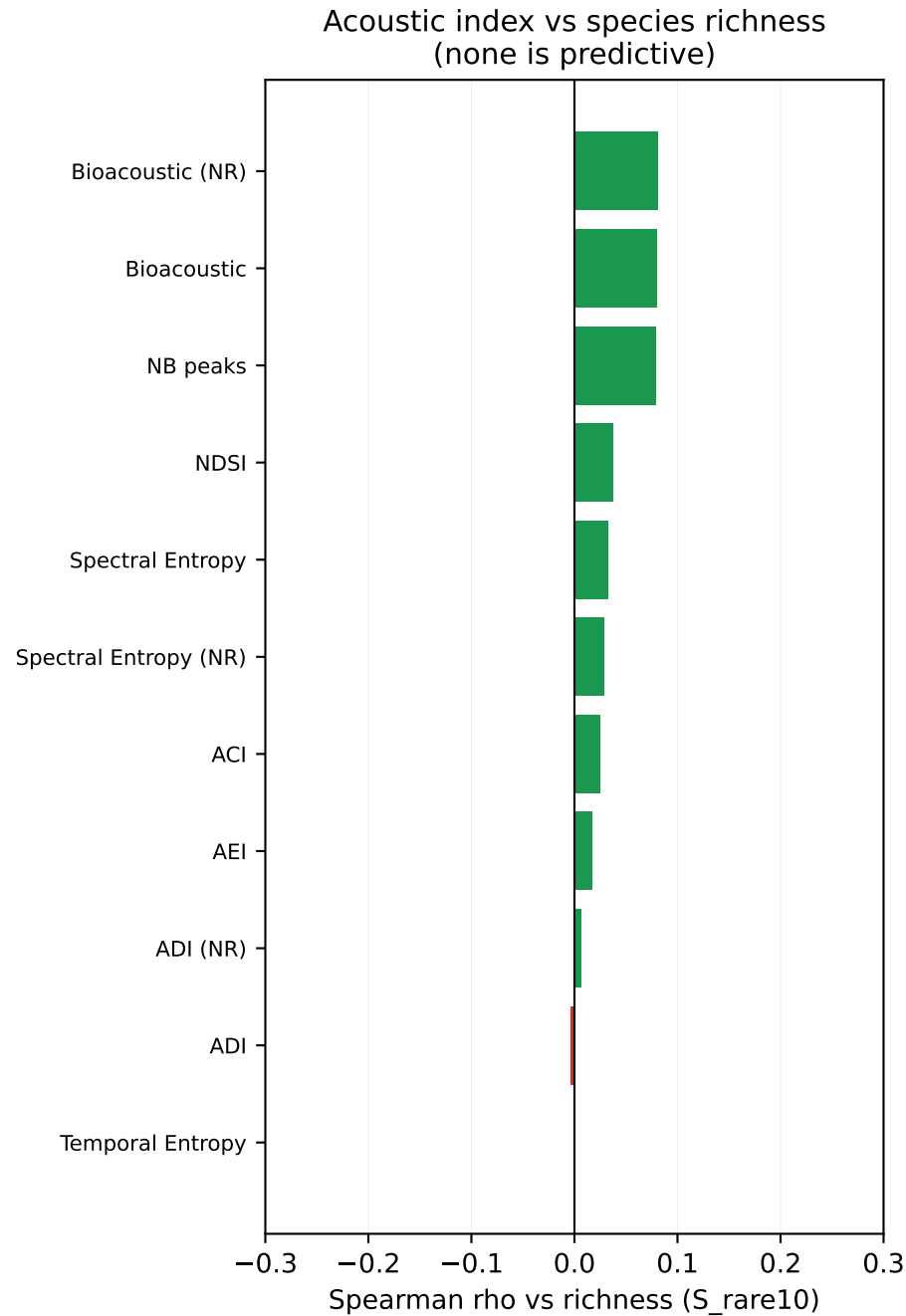
HEADLINE RESULT

- 14,426 cells had enough recordings (≥ 10) to score confidently.
- The acoustic indices did NOT predict species richness (best |Spearman rho| ~ 0.08). The defensible biodiversity metric is therefore effort-controlled species richness, not any acoustic index -- the indices are unreliable on targeted (non-soundscape) recordings that dominate the archive.
- Final labels: 4802 good, 4809 moderate, 4815 bad.

Region-relative biodiversity label per 0.1-degree cell (14,426 well-sampled cells)



Validation: does the audio carry the biodiversity signal?



The Metric & Results

METRIC DEFINITION

location = 0.1 deg x 0.1 deg grid cell (~11 km)
biodiversity value = S_{rare10} , the expected number of distinct recorded species in 10 recordings (Hurlbert sample-based rarefaction; controls for recording effort)
label = tertile of S_{rare10} WITHIN the cell's 10-deg latitude band: good = top third, bad = bottom third
scope = cells with ≥ 10 recordings (14,426 of 49,638 cells)

GLOBAL VALUE RANGE (S_{rare10})

min 1.0 p25 8.4 median 10.3 p75 13.5 p90 17.2 max 80

Median richness by continent (well-sampled cells)

Continent	Median richness	Scored cells
america	11.1	5384
europa	10.3	5202
australia	10.3	459
africa	9.9	1221
asia	9.3	2160

KEY LIMITATIONS (must accompany any use of this metric)

1. Acoustic indices are non-predictive here. ACI/ADI/NDISI etc. were built for passive soundscape monitoring; 68% of archive clips (median length 27 s) are single-target recordings, so per-clip indices do not reflect site diversity.
2. 'Richness' = RECORDED species, not true species. It reflects recordist effort and interest. Rarefaction controls sample SIZE but not observer bias.
3. Weak latitude gradient. Effort-controlled richness is nearly flat across latitude (thresholds ~10-13 everywhere) -- evidence the signal is effort-driven, not the true tropical biodiversity peak. Region-relative labelling is used so scores mean 'rich for its region'.
4. Coverage. Only 29% of cells are confidently scored; the rest are 'insufficient_data' (< 10 recordings).

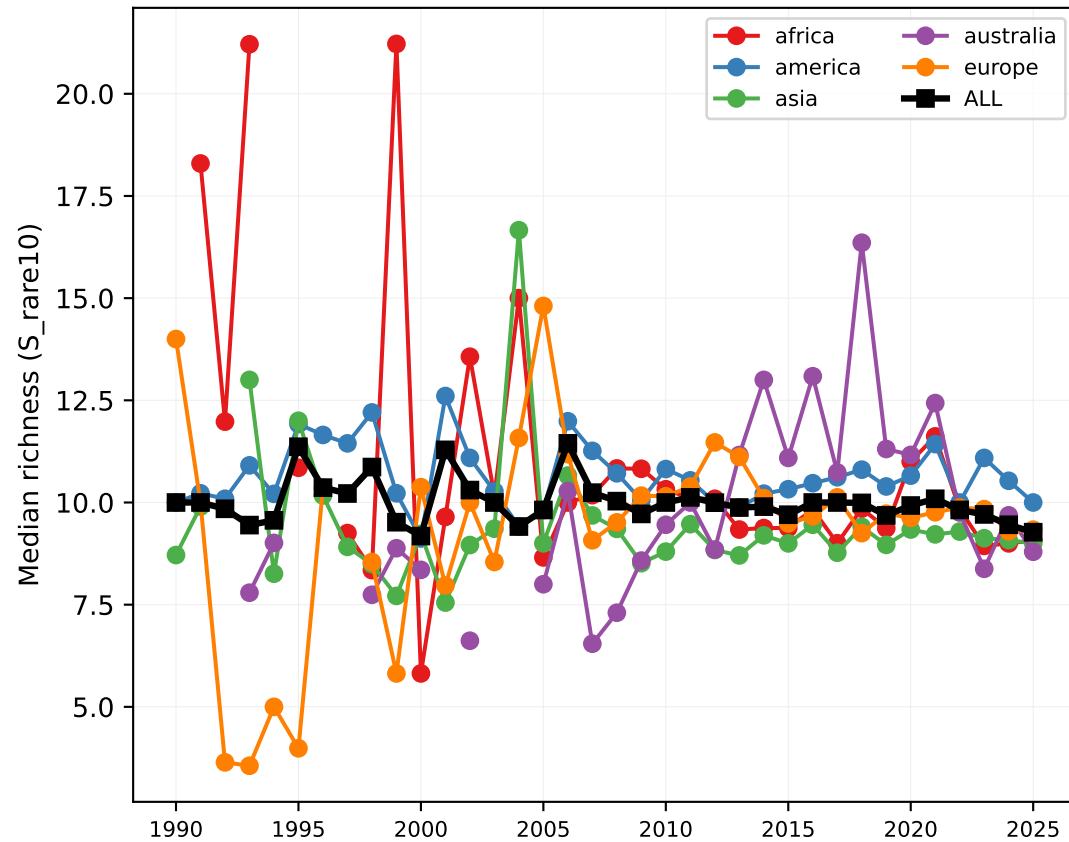
OUTPUT FILES

grid_cells.csv	- per-cell features + richness (all 49,638 cells)
grid_cells_labeled_regional.csv	- region-relative good/moderate/bad labels
biodiversity_map.csv	- lat, lon, label, code -> ready for GIS/folium

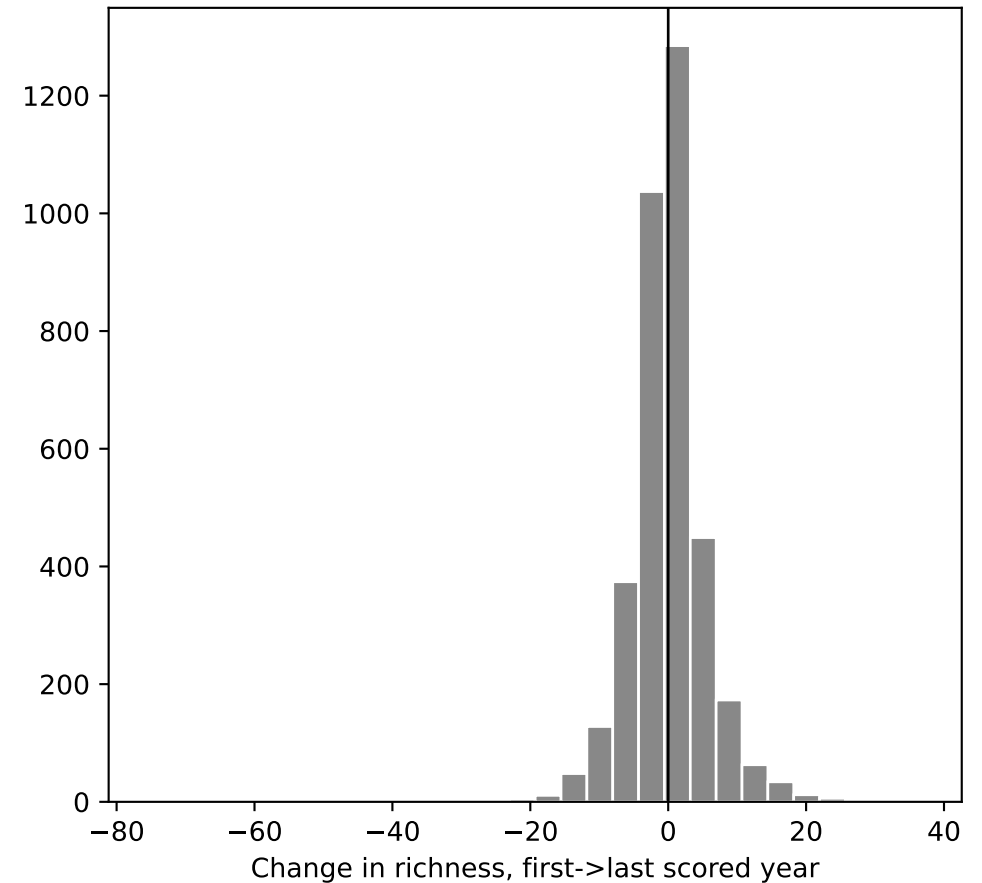
Year-by-year (1990-2025): temporal slices of the metric

years with ≥ 20 scored cells; per-cell change uses every year

Recorded-richness trend by continent



Per-cell change (3637 cells tracked)
up=1578 down=1632 flat=427



COVERAGE: 19,910 scored cell-years; 3637 cells scored in ≥ 2 years, 0 in all 36.

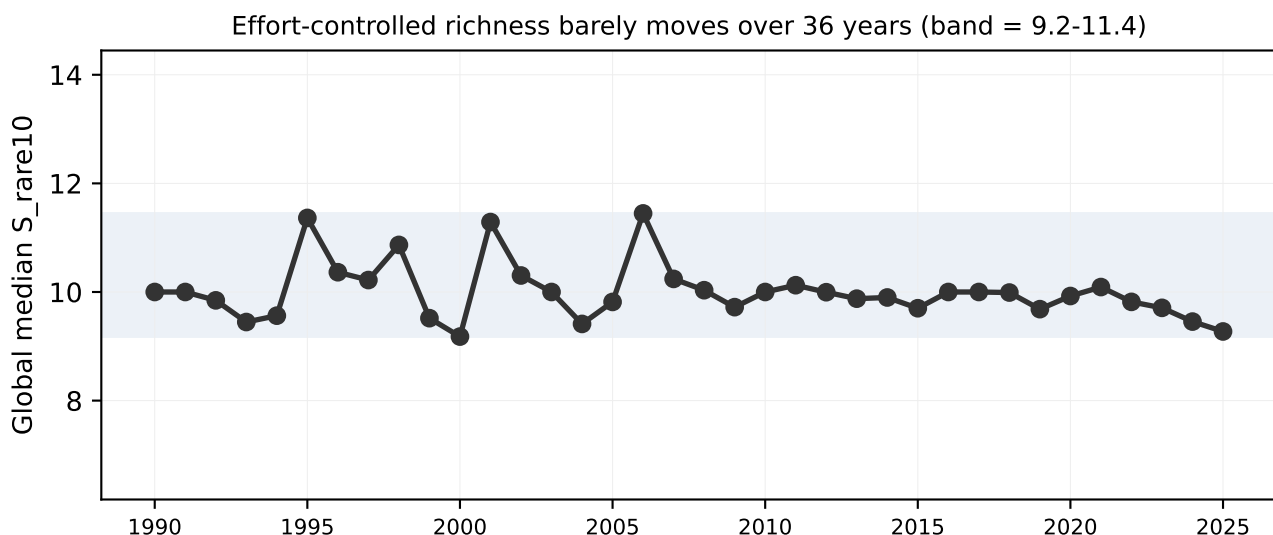
CAUTION: even 36 years of data cannot show real biodiversity change. These movements reflect WHICH cells were recorded and by WHOM each year (effort + observer turnover), not ecological gain or loss. The global change (10.0 $\rightarrow$ 9.3) tracks recording effort, not nature. Use year slices to study sampling coverage over time -- not as a biodiversity time series.

OUTPUTS: grid_cells_yearly.csv (per cell-year), cell_change.csv (per tracked cell).

Long-Run Findings (1990-2025)

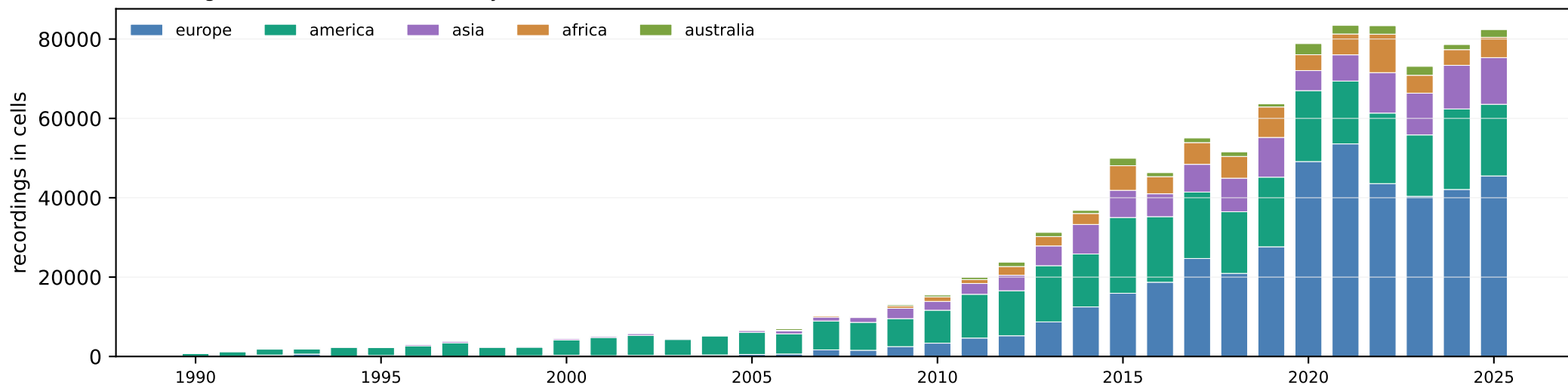
What the multi-decade data adds to the analysis

1. MORE DATA MADE THE ACOUSTIC INDICES LOOK WORSE, NOT BETTER.
Expanding from the 2023-2025 subset to the full 1886-2025 archive (758,198 pipeline-scored recordings, 12,678 scored cells with index means) shrank the best index correlation from $|\rho| \sim 0.16$ to ~ 0.08 . A real signal sharpens with more data; this faded toward zero. (Index means use only recordings processed by this pipeline; the 245,246 merged older ones add richness only.)
2. 36 YEARS OF EFFORT-CONTROLLED RICHNESS ARE ESSENTIALLY FLAT.
Global median S_{rare10} stays within 9.2-11.4 across all 36 years (1990->2025: 10.0 -> 9.3). This reframes the metric as a sampling-effort measure, not ecology. The early years swing most, on the fewest scored cells.
3. TEMPORAL CHANGE IS A COIN-FLIP (RANDOM-WALK SIGNATURE).
Of 3,637 cells tracked across ≥ 2 years, 1,578 rose and 1,632 fell (flat=427). That near-perfect symmetry over 36 years is the fingerprint of observer turnover and noise, not directional change.
4. SPATIALLY RICH, BUT TEMPORALLY STILL STARVED.
Coverage grew to 49,638 cells / 14,426 scored -- a much denser map -- yet only 0 cells were scored in all 36 years (3,620 in ≥ 2 of them). The biodiversity MAP is far stronger; a true TIME SERIES remains impossible from this data.

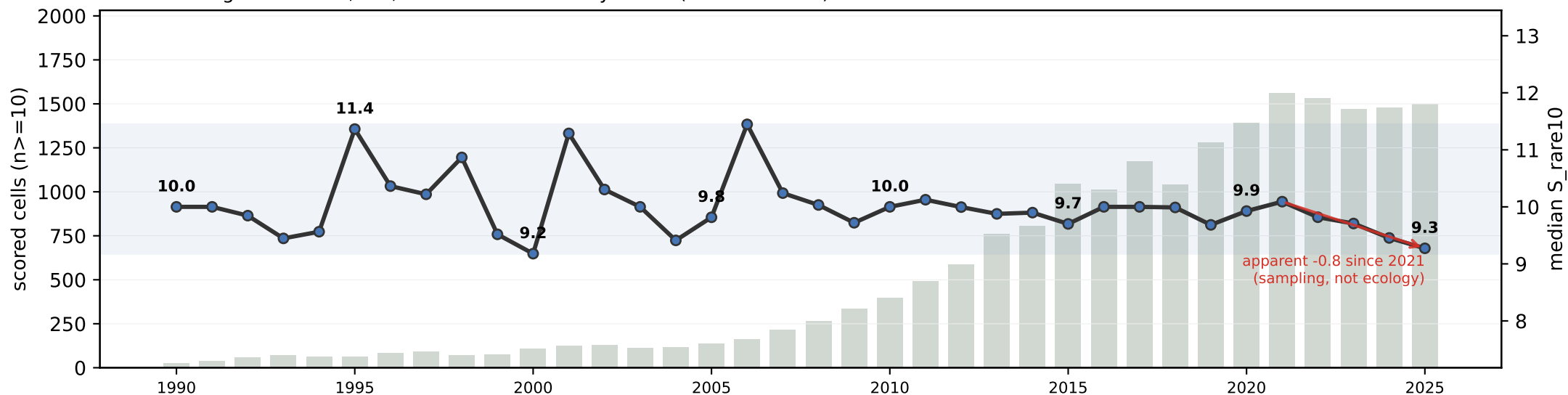


Year-by-year: effort vs. the metric (1990-2025)

Recording effort behind the metric, by continent (977 -> 82,378, +8331%)



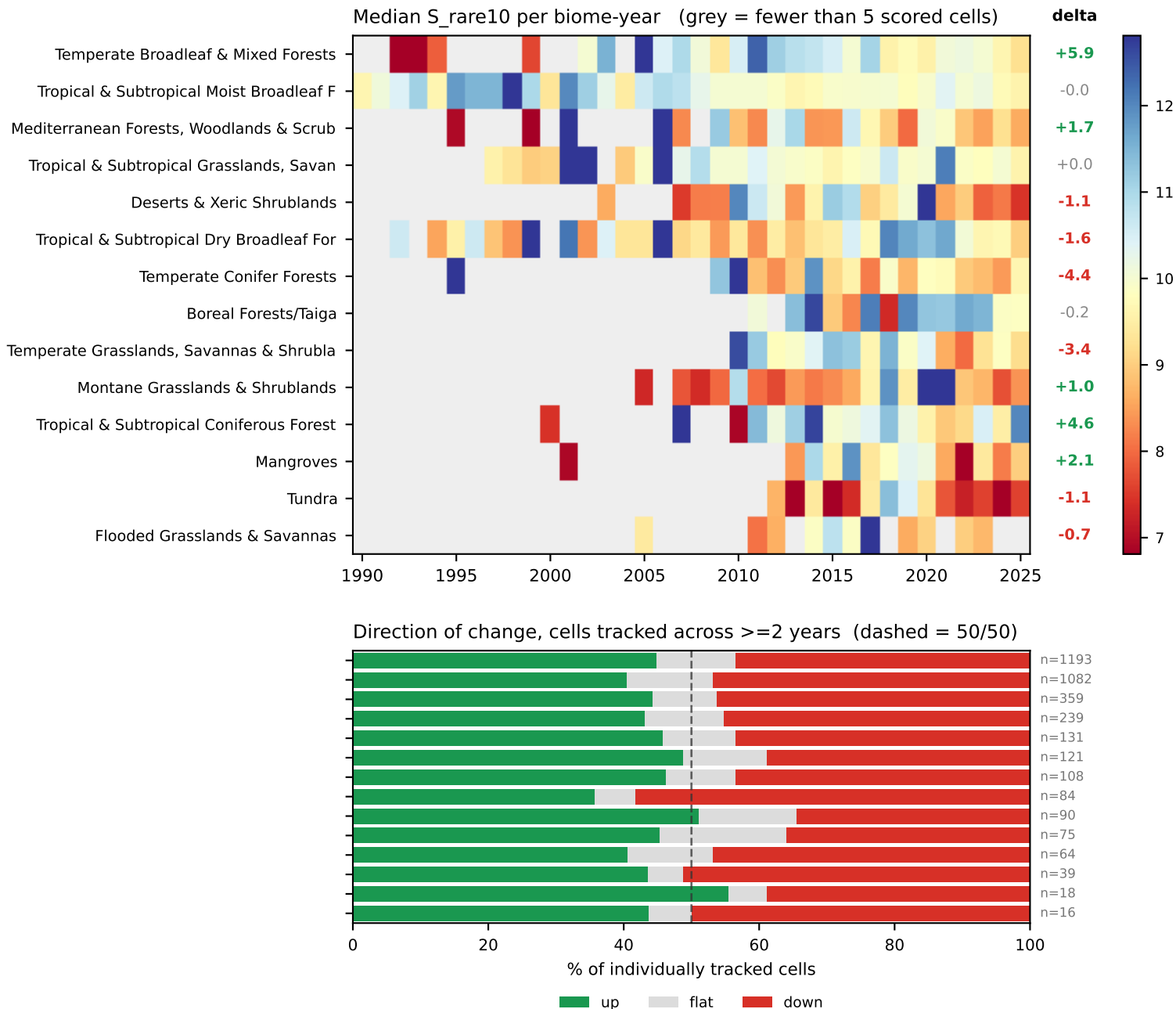
Scored cells grew 25 -> 1,501; median richness stayed flat (9.2-11.4 band)



Bars = recording EFFORT (geolocated recordings entering the metric); line = effort-controlled median richness. Effort rose 8,331% (1990-2025) while the metric stayed in a band. The apparent dip (red arrow) is NOT ecological degradation: it tracks recent-upload completeness and shifting recordist coverage. Do not read the decline as biodiversity loss.

Year-by-Year Change by Biome (1990-2025)

Effort-controlled richness (*S_rare10*) split by terrestrial biome



NO BIOME SHOWS COHERENT DIRECTIONAL CHANGE.

TESTED YEAR BY YEAR (Kendall tau on each plotted series), 3 of 14 biomes reach $p < 0.05$ -- about the 0.7 expected by chance -- and none survives Bonferroni (smallest $p = 0.009$ vs 0.0036).

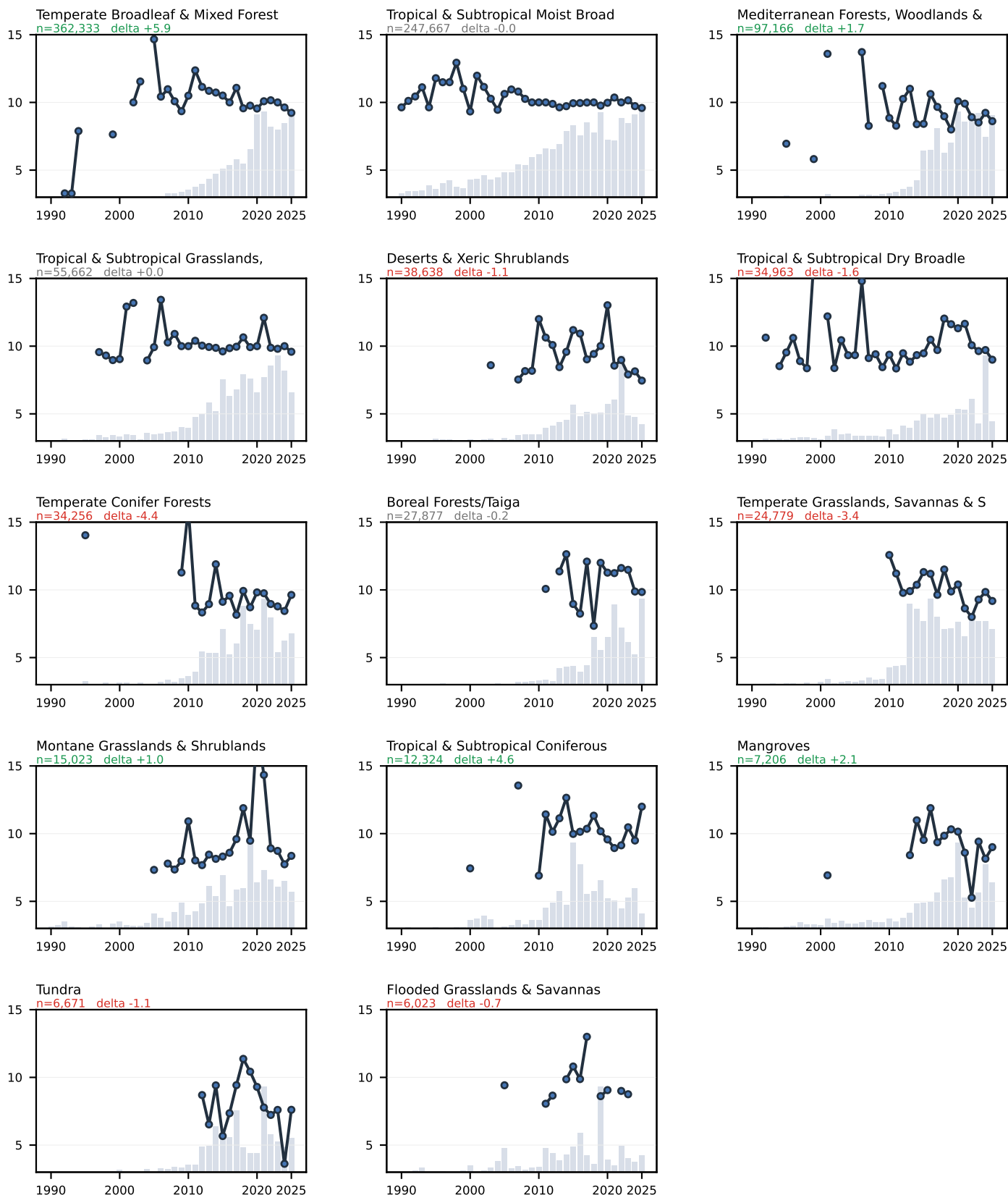
In 12 of 14 biomes one step between consecutive plotted years is at least as large as the first->last delta, and the cells tracked inside each biome split close to 50/50 (1,574 up vs 1,623 down overall). The largest delta -- Temperate Broadleaf & Mixed Forests at +5.9 (1992->2025) -- rests on a series whose largest single step is 4.6. Counted from its first year with ≥ 5 scored cells (1986, as *biome_change.csv* and the Silent Signal page do), the same delta is -0.3: its sign depends on where the series starts.

READ THIS AS SAMPLING, NOT ECOLOGY. *S_rare10* controls for the NUMBER of recordings in a cell, but not for who recorded, for how long, or with what target. A biome's series therefore moves as its recordist community turns over, and it swings hardest where a biome-year rests on few cells: steps touching a year with < 20 scored cells have a median of 1.3 (max 7.9); steps between years with ≥ 100 cells, 0.3 (max 2.2). That is the signature of small samples rather than of habitat change.

METHOD. Each 0.1-degree cell takes the modal biome of the recordings inside it. 99.3% of recordings fall in their cell's modal biome and only 1.8% of cells span more than one, so boundary cells cannot drive these patterns. Biomes are ordered by recording volume.

Each Biome, Year by Year (1990-2025)

Line = median S_{rare10} (left axis). Bars = recordings entering the metric (right axis).

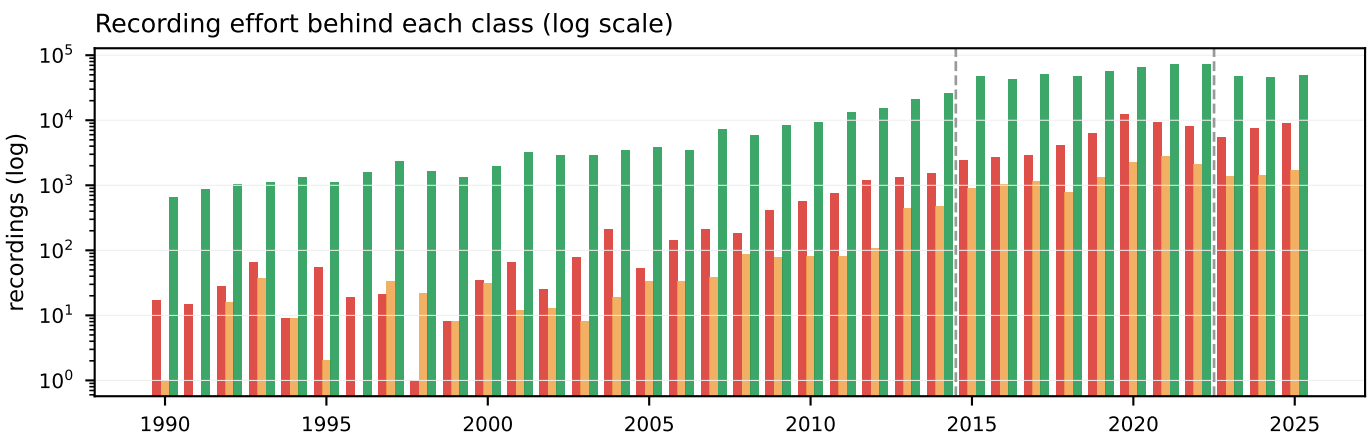
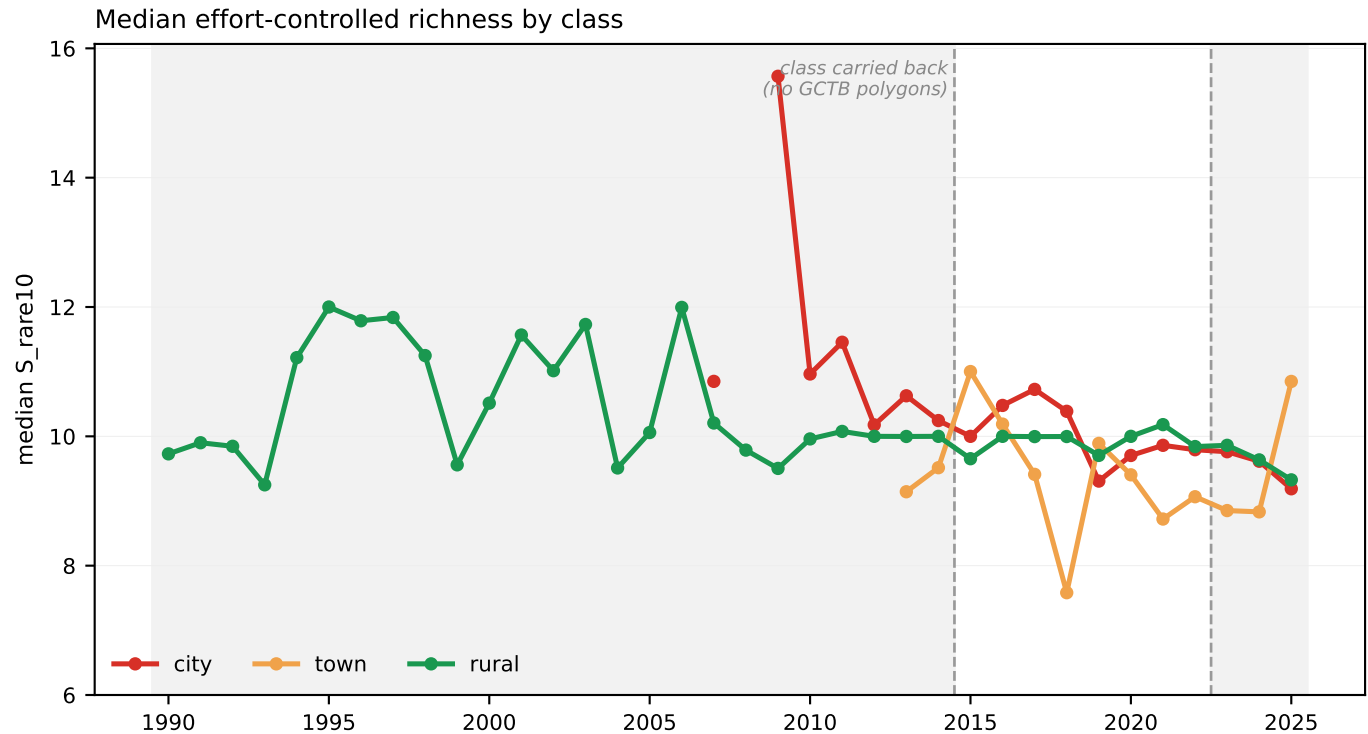


Every panel is drawn on the SAME richness axis (3-15), so the series can be compared directly. Read the grey bars first: where effort is small the line is jagged, and where it is large the line is flat -- most clearly within one biome. Temperate Broadleaf (362,333 recordings) spans 3.3-14.7 over the chart, but only 9.2-12.4 across its 15 years with ≥ 100 scored cells. 10 of 14 biomes never reach 100 scored cells in a year, and their lines stay jagged throughout.

No panel shows a monotonic long-run trend. The deltas printed above each panel are first-vs-last plotted-year differences, not fitted trends; in 12 of 14 biomes a single step between consecutive plotted years is at least as large as the delta, so a delta alone does not separate change from noise.

Year-by-Year Change by Urban Class (1990-2025)

Effort-controlled richness (S_{rare10}) for city / town / rural cells



NEITHER CITY NOR RURAL RICHNESS TRENDS.

Over the labelled window (2015-2022) the city median moves -0.2 and rural $+0.2$. Both are far smaller than the year-to-year jitter in either series, so neither is distinguishable from noise. The town series (only 720 cells) swings by ~ 3 species between adjacent years, which is what a small sample looks like.

THE TRACKED CELLS SAY NO CHANGE. Cells followed across ≥ 2 years split city 123/119 up/down and rural 1318/1365 -- rural is an exact coin flip. A real divergence between urban and rural biodiversity would show up here first, and it does not.

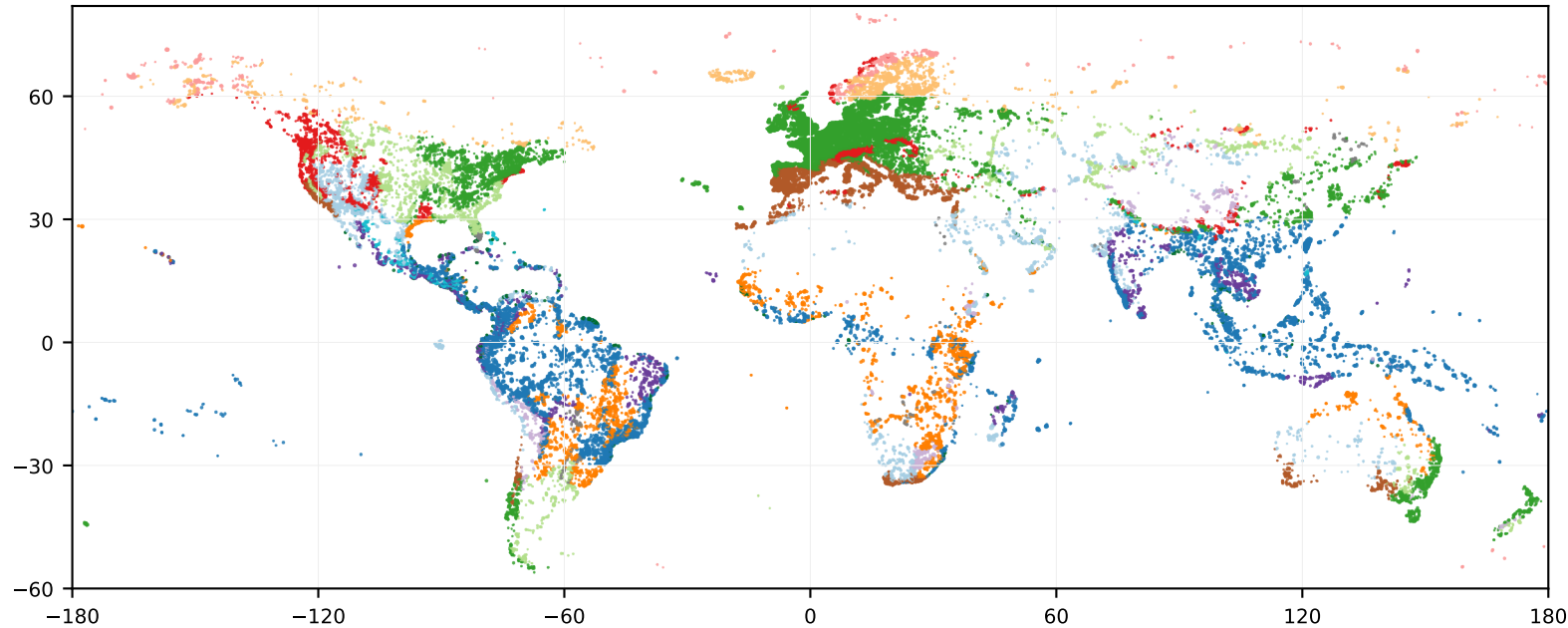
COVERAGE CAVEAT. GCTB built-up polygons cover 2015-2022 only, so only 512,183 recordings (2015-2022) carry a real class. Urban class is a property of the PLACE, so each cell's class is carried back to 1990 and forward to 2025 (shaded). Those years rest on an assumption -- that cells did not change built-up status -- not on measurement, and it grows weaker the further back it is carried.

Cells take the modal class of their recordings: 97.0% of recordings fall in their cell's modal class and 5.8% of cells are mixed (higher than the 1.8% for biomes, because city boundaries cut through 0.1-degree cells far more often than biomes do).

Recording Coverage by Biome

49,344 grid cells (0.1 deg), each coloured by the modal biome of its recordings

Dot size = recordings in that cell (log scale)



biome (cells / recordings)

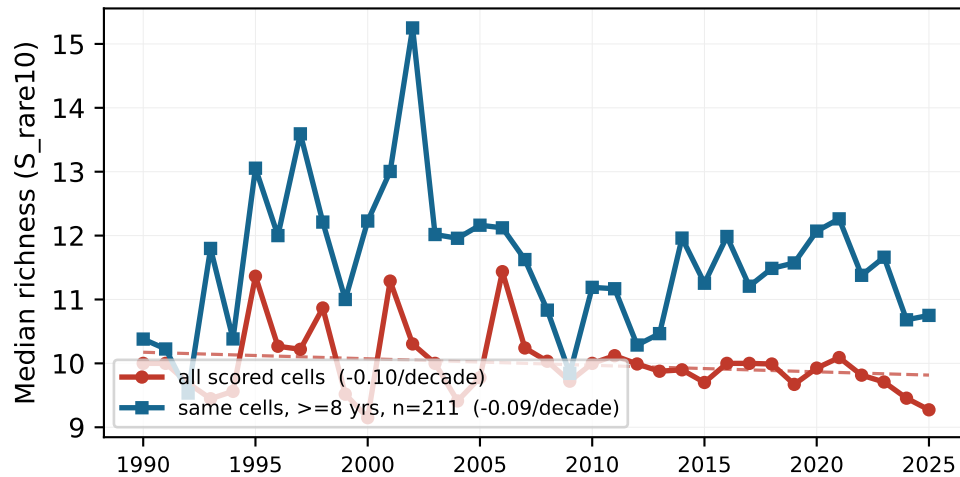
- Temperate Broadleaf & Mixed Forests
15,729 cells / 361,583 rec
- Tropical & Subtropical Moist Broadleaf
9,569 cells / 248,736 rec
- Mediterranean Forests, Woodlands & Scr
4,715 cells / 97,199 rec
- Tropical & Subtropical Grasslands, Sav
3,421 cells / 55,755 rec
- Deserts & Xeric Shrublands
2,951 cells / 38,677 rec
- Tropical & Subtropical Dry Broadleaf F
1,762 cells / 35,123 rec
- Temperate Conifer Forests
3,185 cells / 34,264 rec
- Boreal Forests/Taiga
2,104 cells / 27,879 rec
- Temperate Grasslands, Savannas & Shrub
2,734 cells / 24,833 rec
- Montane Grasslands & Shrublands
1,091 cells / 15,060 rec
- Tropical & Subtropical Coniferous Fore
552 cells / 12,364 rec
- Mangroves
459 cells / 7,249 rec
- Tundra
725 cells / 6,666 rec
- Flooded Grasslands & Savannas
347 cells / 6,037 rec

COVERAGE IS NOT ECOLOGICAL COVERAGE. The map shows where recordists go, not where biomes are. Europe is saturated across a single biome band while whole tropical biomes are covered by scattered cells: Temperate Broadleaf & Mixed Forests holds 15,729 cells, the three least-recorded together only 1,531. This is the sampling bias behind every per-biome number here -- a biome comparison is a comparison of recordist communities as much as of habitats, and the biome-years resting on the fewest cells are the ones whose series swing hardest on pages 8 and 9.

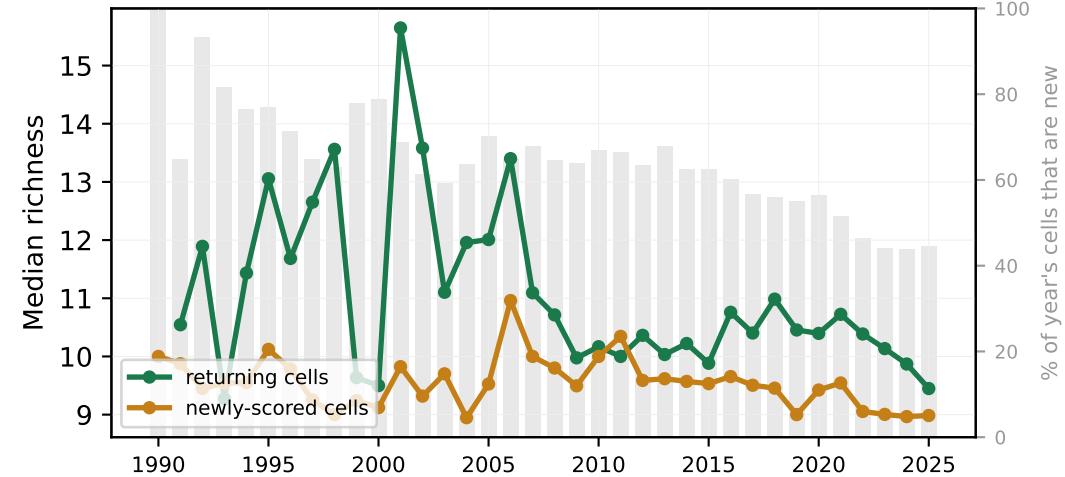
Cells take the modal biome of their recordings; 883 (1.8%) contain more than one and are drawn in their majority colour. Polygons: RESOLVE Ecoregions 2017 (Dinerstein et al., CC-BY 4.0). Colours are categorical only.

Is there a long-run trend (1990-2025)? Turnover, not change

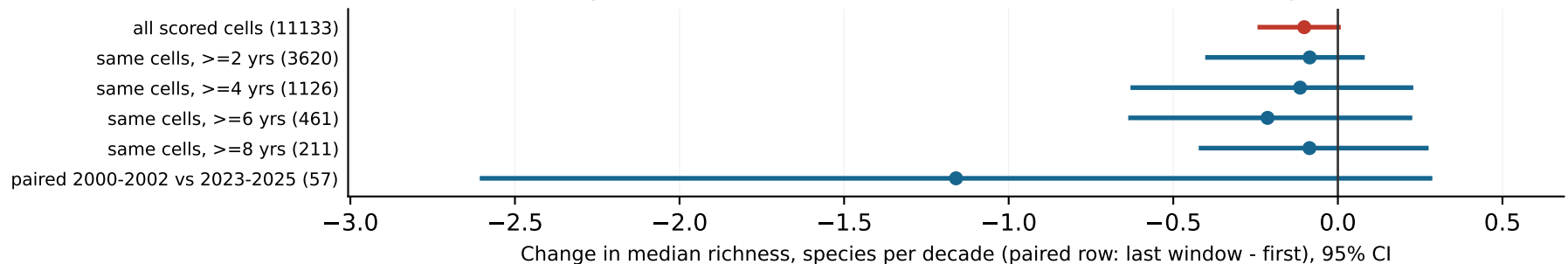
The decline is in the pool, not the places



Newly-scored cells are poorer in 33 of 35 years



Every turnover-controlled estimate sits on zero -- but none is precise



NO SIGNIFICANT TREND. Raw, the series falls 10.00 -> 9.27 (Theil-Sen -0.10 species/decade, Kendall tau -0.20, $p = 0.093$) -- not significant at 0.05. Follow the SAME cells: the >=8-year panel gives -0.09/decade and the 57 cells paired 2000-2002 vs 2023-2025 change by -1.160 (23 up / 34 down).

MECHANISM: 44-100% of each year's scored cells were never scored before, and newly-scored cells sit below returning ones in 33 of 35 years. The archive keeps expanding into thinner locations, which drags the pooled median down while individual places move up and down in roughly equal numbers.

LIMIT: the controlled estimates are UNDERPOWERED, not proof of stability. The paired CI is [-2.61, +0.29] and the >=8-year panel's is [-0.42, +0.28] per decade, so a real change of that size would not be detectable here. Years before 1990 have < 20 scored cells and are omitted.

CONFOUND: the data source changes with time. Before 2015 nearly every recording comes from the merged external table (merged_metadata.all.csv), 2015-2022 from the historical backfill, 2023-2025 from the base+gap run -- each with its own download filters. Year and provenance cannot be separated with what is on disk, so even the raw trend is suspect.